CoRide: Joint Order Dispatching and Fleet Management for Multi-Scale Ride-Hailing Platforms

Jiarui Jin[1], Ming Zhou[1], Weinan Zhang[1], Minne Li[2], Zilong Guo[1], Zhiwei Qin[3], Yan Jiao[3], Xiaocheng Tang[3], Chenxi Wang[3], Jun Wang[2], Guobin Wu[4], Jieping Ye[3]

[1]Shanghai Jiao Tong University, [2]University College London, [3]DiDi AI-Labs, [4]DiDi Research

Slides: https://jinjiarui.github.io/files/cikm19.pdf

Content

- Problem Background
  - Ride-hailing Platform
  - Multi-scale Joint Ride-hailing Platform
- Architecture
  - Hierarchical Multi-agent Reinforcement Learning
- Experiment
  - Simulator
  - Real-world Data Experiments
  - Synthetic Data Experiments
- Conclusion

Content

- **Problem Background**
  - Ride-hailing Platform
  - Multi-scale Joint Ride-hailing Platform
- Architecture
  - Hierarchical Multi-agent Reinforcement Learning
- Experiment
  - Simulator
  - Real-world Data Experiments
  - Synthetic Data Experiments
- Conclusion

Ride-hailing Platform

- What RL
  - order dispatching
  - fleet management

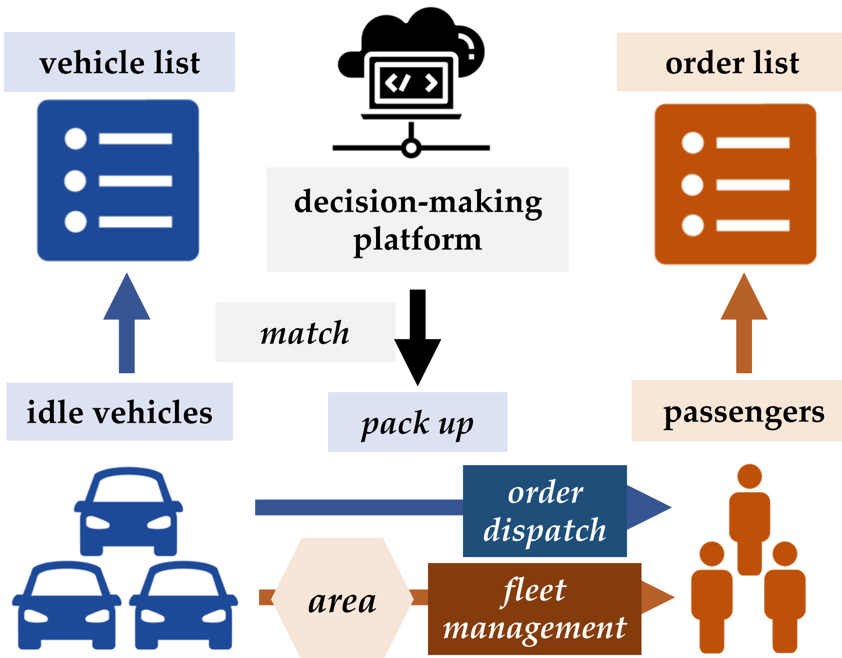


Figure: Architecture of Ride-hailing

Ride-hailing Platform

- What RL
  - order dispatching
  - fleet management
- Why RL
  - sequential decision making process
  - dynamic real-world scenario

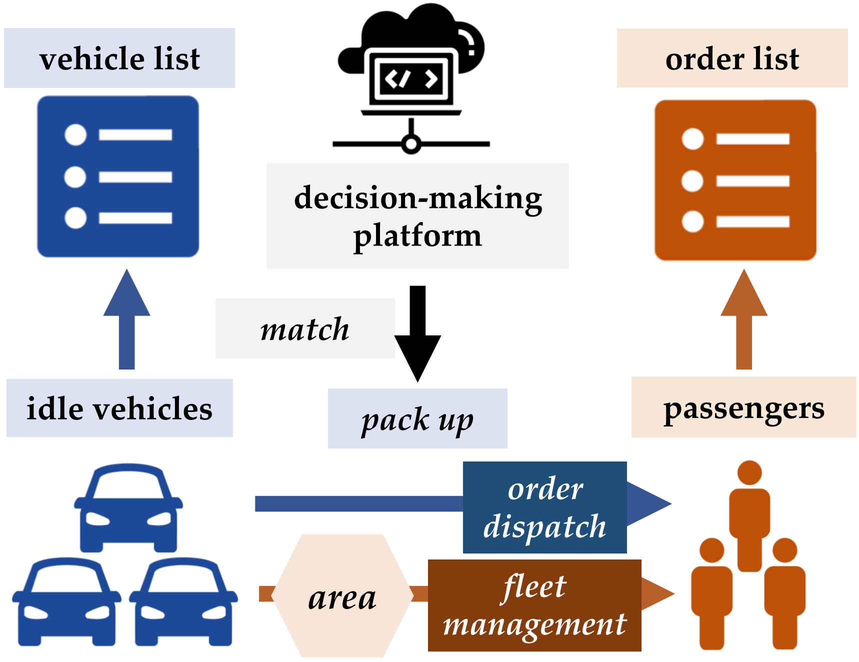


Figure: Architecture of Ride-hailing

Ride-hailing Platform

- What RL
  - order dispatching
  - fleet management
- Why RL
  - sequential decision making process
  - dynamic real-world scenario
- How RL

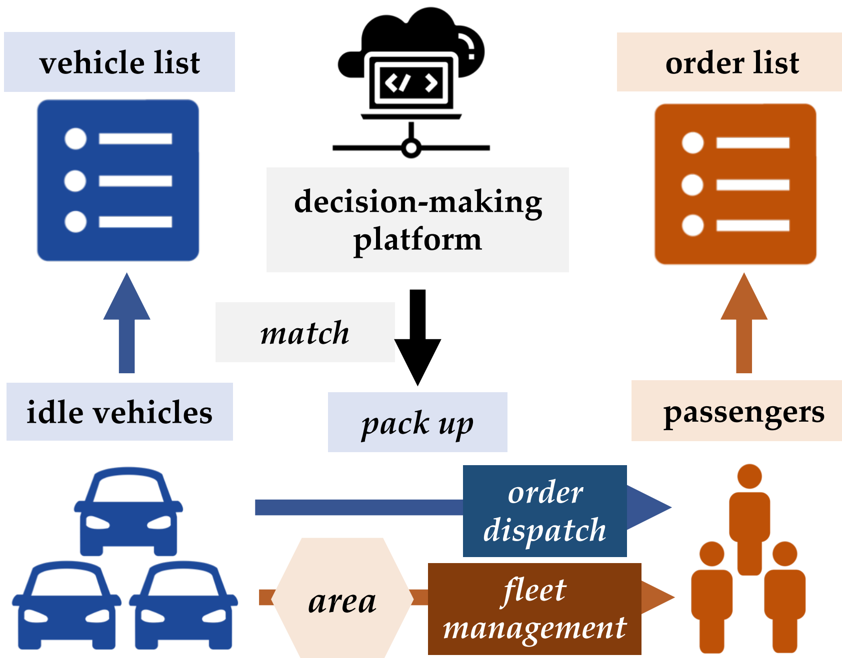


Figure: Architecture of Ride-hailing

Joint Multi-Scale Ride-hailing Platform

- Large-scale agent
  regard **vehicle** as agent

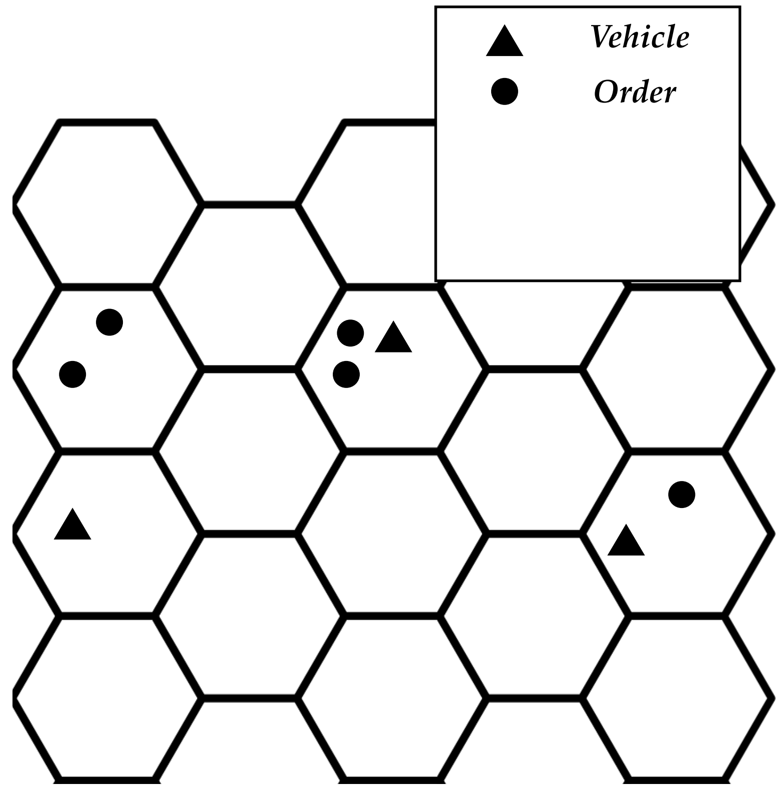


Figure: Grid World of Ride-hailing

Ref: Minne Li, et al. Efficient Ride-sharing Order Dispatching with Mean Field Multi-agent Reinforcement Learning. WWW, 2019.

Joint Multi-Scale Ride-hailing Platform

- Large-scale agent
  regard **vehicle** as agent

- Immediate & future reward
  combine immediate gain with future value in the **one** time credit

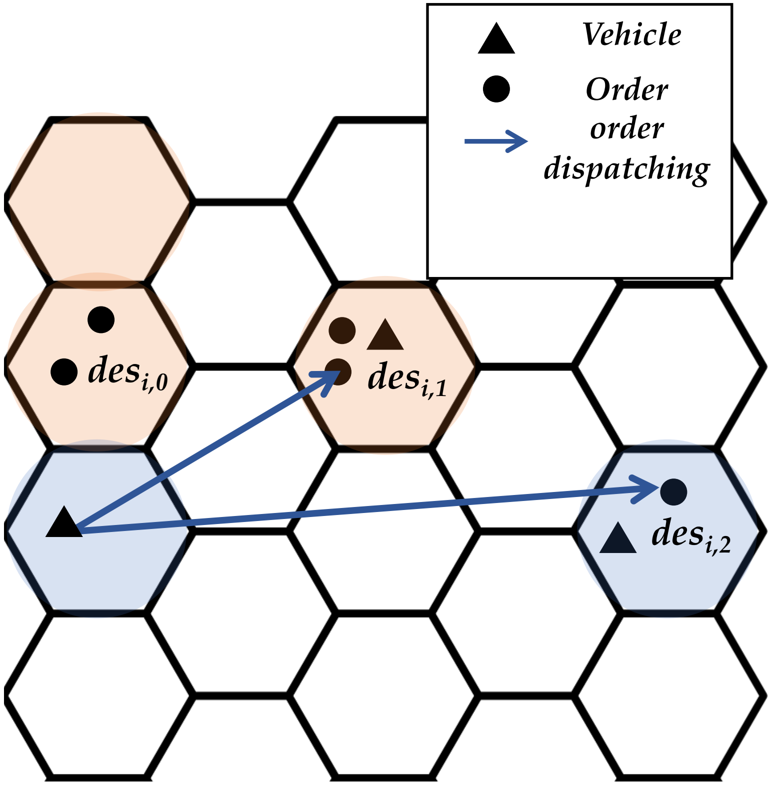


Figure: Grid World of Ride-hailing

Ref: Zhiwei Qin, et al. Deep Reinforcement Learning with Applications in Transportation. Tutorial at KDD, 2019.

Joint Multi-Scale Ride-hailing Platform

- Large-scale agent
  regard **vehicle** as agent
- Immediate & future reward
  combine immediate gain with future value in the **one** time credit
- Heterogeneous & variant action space
  separately study order dispatching and fleet management

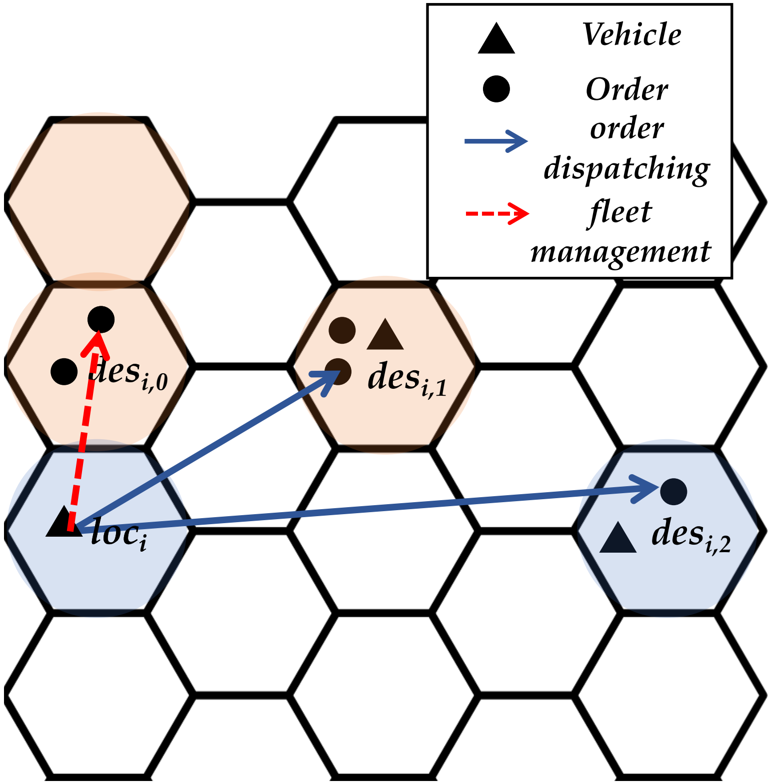


Figure: Grid World of Ride-hailing

Ref: Zhiwei Qin, et al. Deep Reinforcement Learning with Applications in Transportation. Tutorial at KDD, 2019.

University College London
DiDi AI LABS
Apex Data & Knowledge Management Lab

Joint Multi-Scale Ride-hailing Platform

- Large-scale agent

  regard **vehicle** as agent

- Immediate & future reward

  combine immediate gain with future value in the **one** time credit

- Heterogeneous & variant action space

  separately study order dispatching and fleet management

- Multi-scale ride-hailing

  treat grid in the **same** level.

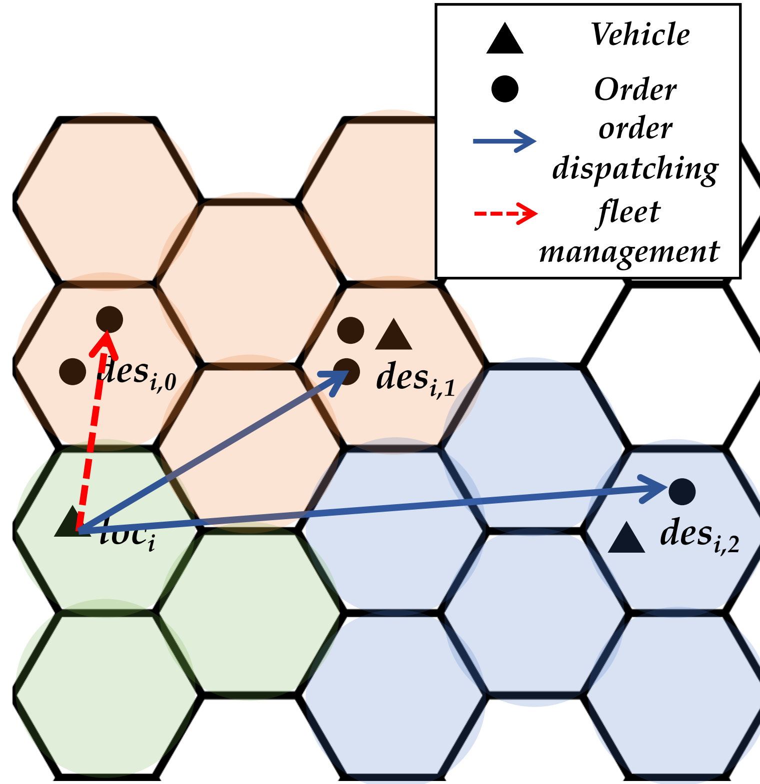


Figure: Grid World of Ride-hailing

Ref: Zhe Xu, et al. Large Scale Order Dispatching in On Demand Ride Hailing Platforms: A learning and Planning Approach. KDD, 2018.

Joint Multi-Scale Ride-hailing Platform

- Large-scale agent

  regard **grid (vehicle)** as agent

- Immediate & future reward

  combine immediate gain with future value in the **different (same)** time credit

- Heterogeneous & variant action space

  jointly (**separately**) study order dispatching and fleet management

- Multi-scale ride-hailing

  treat grid in the **different (same)** level.

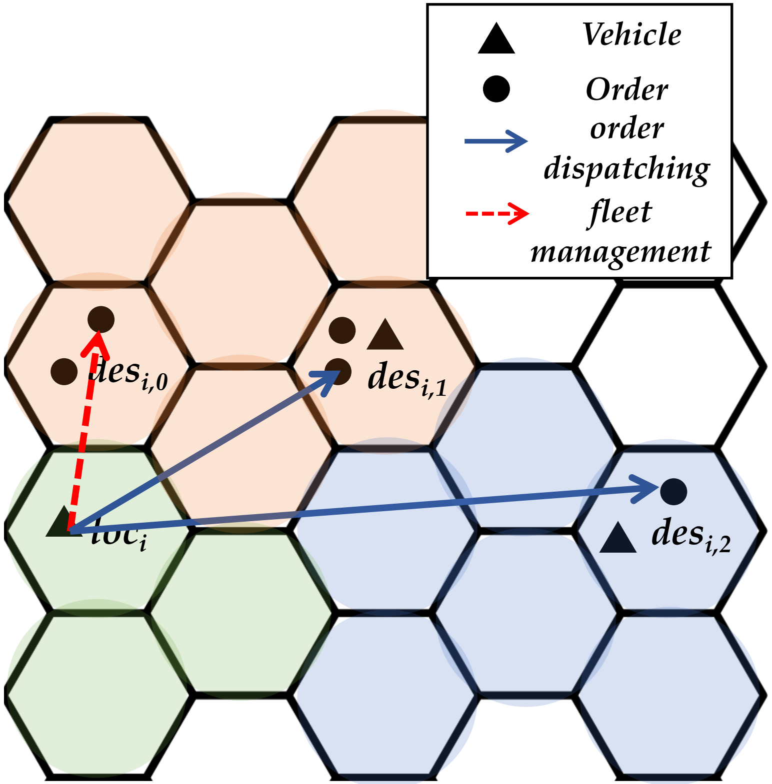


Figure: Grid World of Ride-hailing

Joint Multi-Scale Ride-hailing Platform

Grid World

Figure: Hierarchical Multi-agent Reinforcement Learning

Content

- Problem Background
  - Ride-hailing Platform
  - Multi-scale Joint Ride-hailing Platform
- **Architecture**
  - Hierarchical Multi-agent Reinforcement Learning
- Experiment
  - Simulator
  - Real-world Data Experiments
  - Synthetic Data Experiments
- Conclusion

Hierarchical Multi-agent Reinforcement Learning

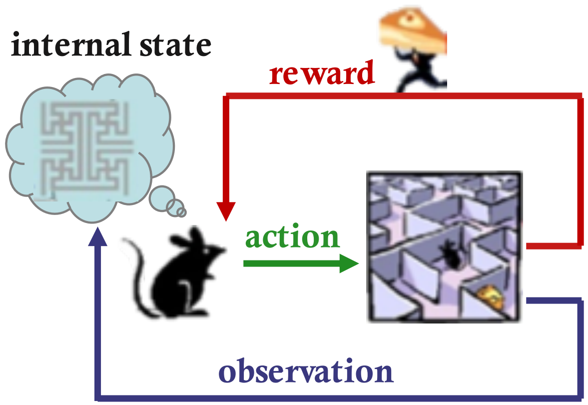


Figure: Single-agent Reinforcement Learning

Hierarchical Multi-agent Reinforcement Learning

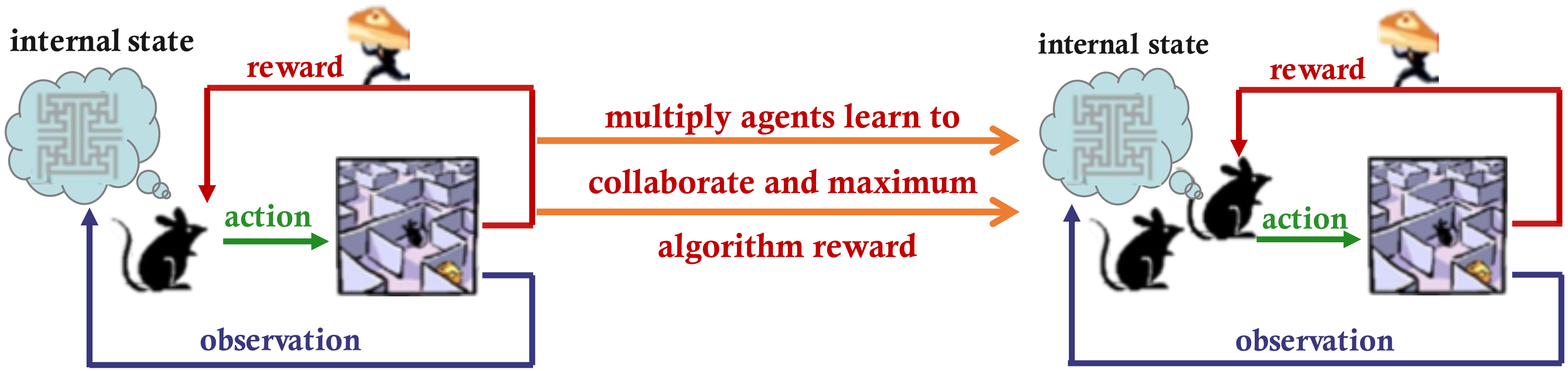


Figure: Single-agent Reinforcement Learning

Figure: Multi-agent Reinforcement Learning

Hierarchical Multi-agent Reinforcement Learning

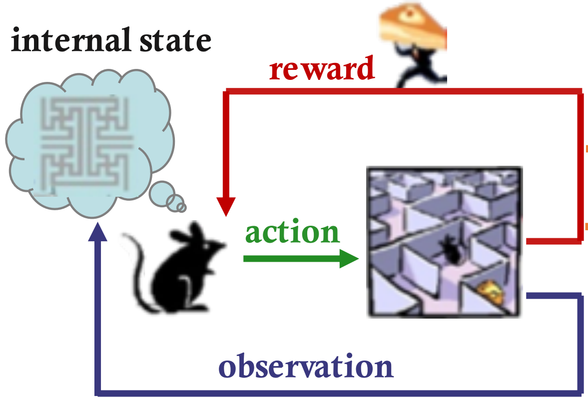


Figure: Single-agent Reinforcement Learning

multiply agents learn to
collaborate and maximum
algorithm reward

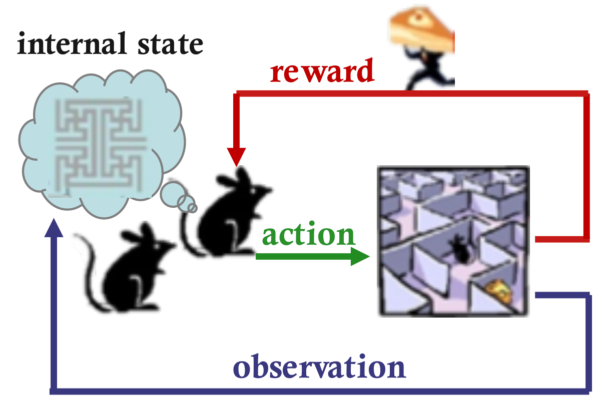


Figure: Multi-agent Reinforcement Learning

Figure: Hierarchical Reinforcement Learning

UCL University College London
DiDi AI LABS
Apex Data & Knowledge Management Lab

Joint Multi-Scale Ride-hailing Platform

Grid World

Figure: Hierarchical Multi-agent Reinforcement Learning

Hierarchical Multi-agent Reinforcement Learning

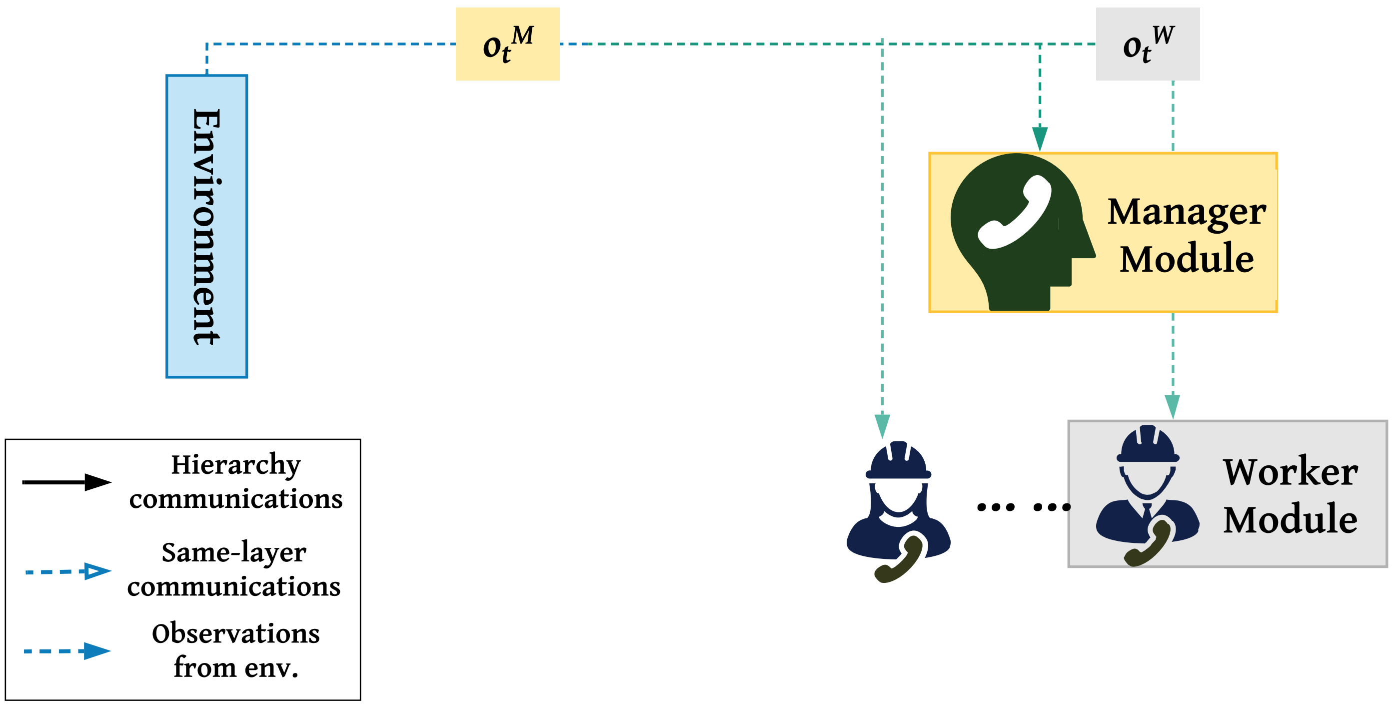


Figure: Hierarchical Reinforcement Learning

Hierarchical Multi-agent Reinforcement Learning

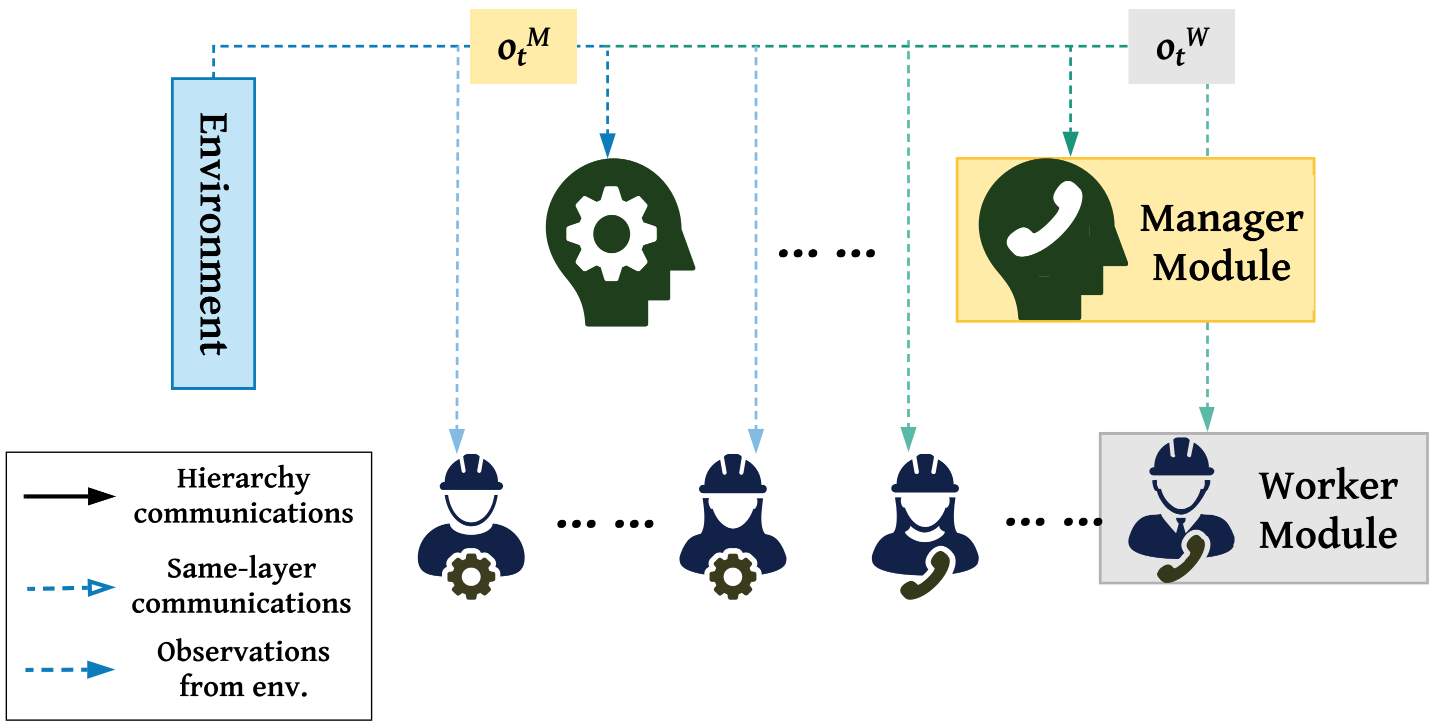


Figure: Hierarchical Multi-agent Reinforcement Learning

Hierarchical Multi-agent Reinforcement Learning

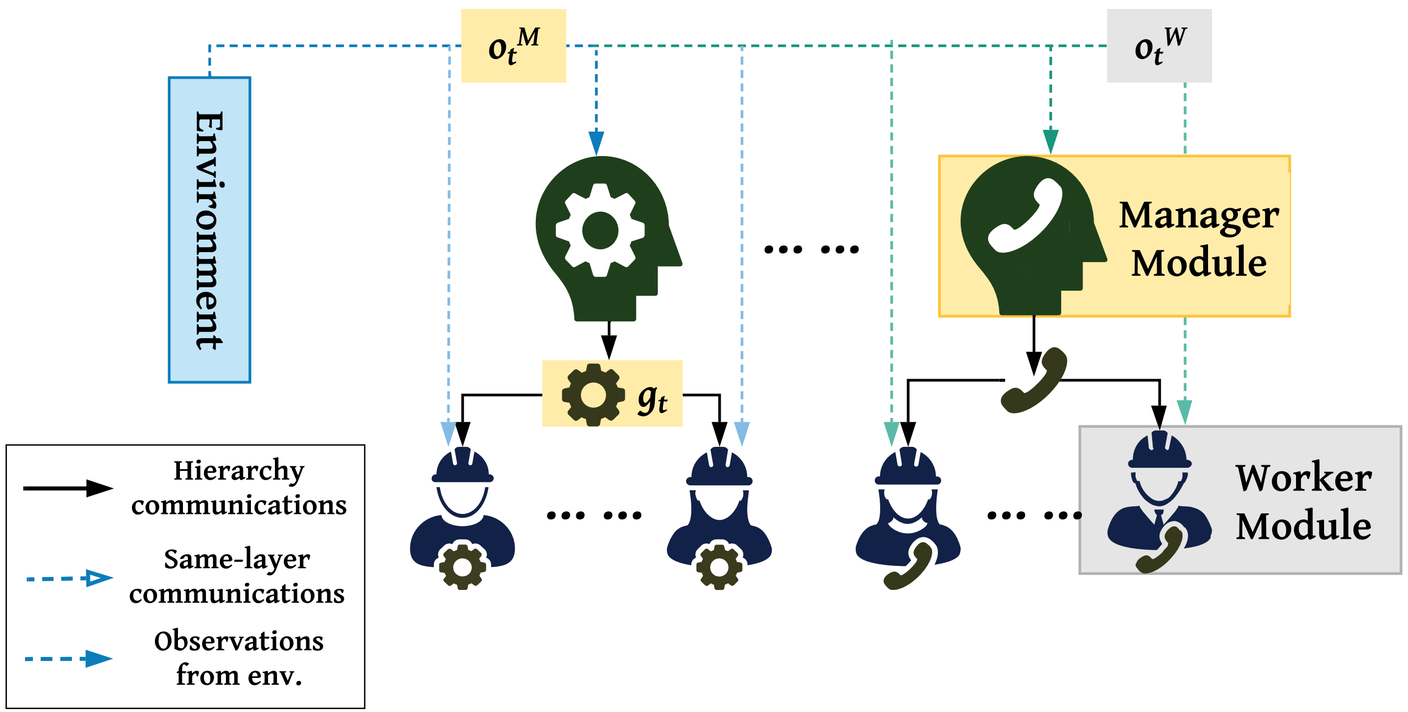


Figure: Hierarchical Multi-agent Reinforcement Learning

Hierarchical Multi-agent Reinforcement Learning

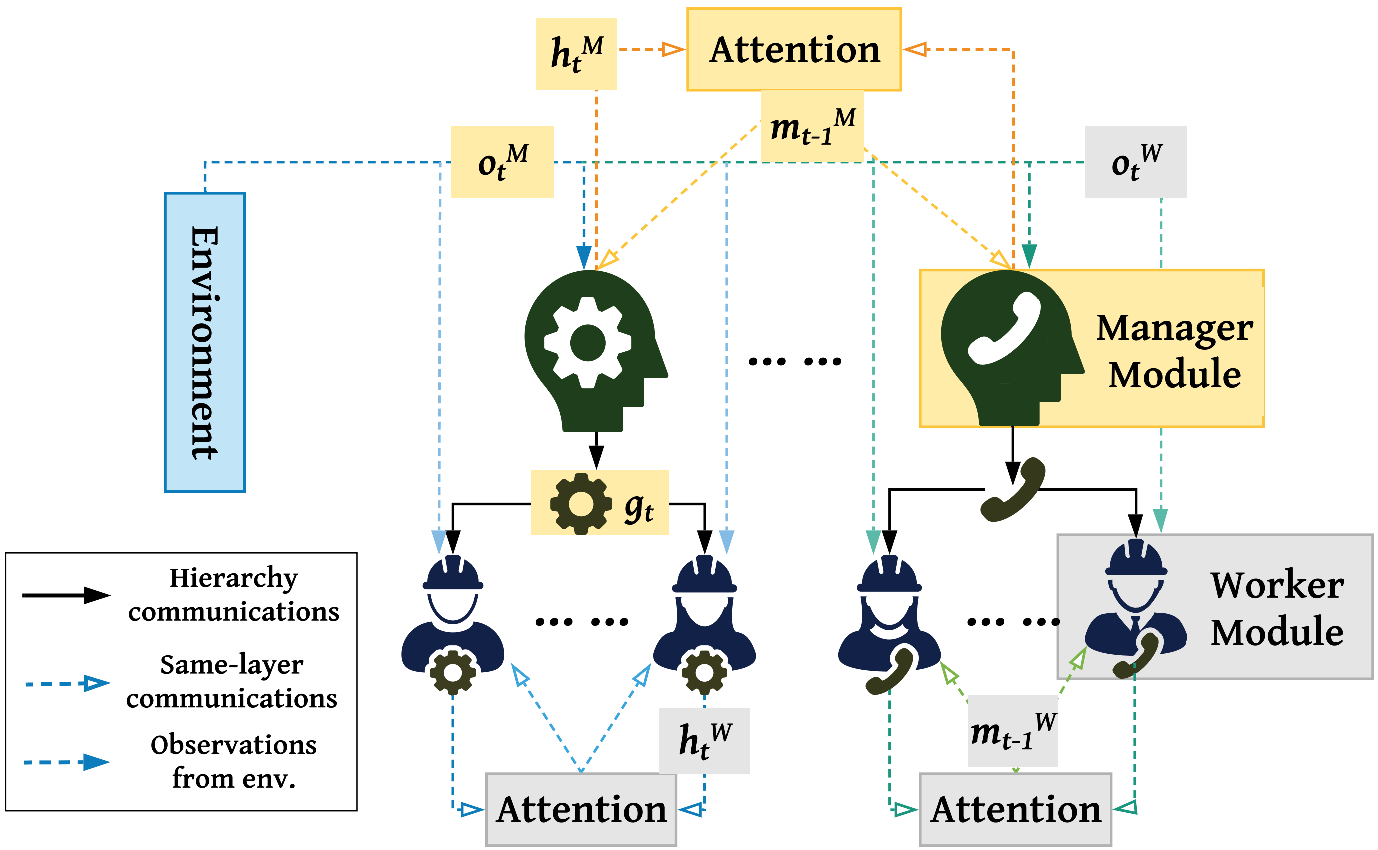


Figure: Hierarchical Multi-agent Reinforcement Learning

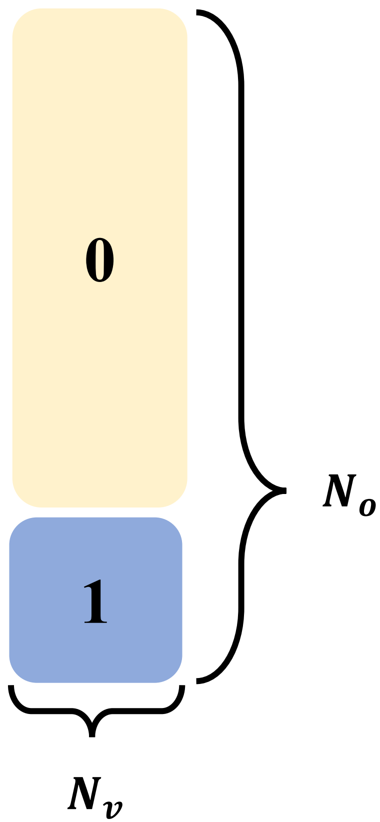
University College London DiDi AI LABS Apex Data & Knowledge Management Lab

Component Modules

- State $s_t \in \mathcal{S}$, Observation $o_t \in \mathcal{O}$:
  - each agent i draws private observation o_t^i correlated with s_t
  - observation of *manager* is joint observation of its *worker*s
  - $\mathcal{S} =< N_v, N_o, \boldsymbol{E}, N_f, D_o >$: number of vehicle, number of order, entropy, number of fleet management vehicle, order distribution

Component Modules

• S



$$E = -k_B \sum_i \rho ln \boldsymbol{\rho}$$
$$= -k_B \times \rho ln \left(\frac{\boldsymbol{N_v} \times \boldsymbol{N_v}}{\boldsymbol{N_v} \times \boldsymbol{N_o}} \right)$$

py,

Possible Pairs: $(\boldsymbol{N_v}, \boldsymbol{N_o})$
Dispatched Pairs: $(\boldsymbol{N_v}, \boldsymbol{N_v})$

conditioned on $\boldsymbol{N_o} > \boldsymbol{N_v}$, and straightforward to transform to other situations

Component Modules

- State $s_t \in \mathcal{S}$, Observation $o_t \in \mathcal{O}$:
  - each agent i draws private observation o_t^i correlated with s_t
  - observation of *manager* is joint observation of its *worker*s
  - $\mathcal{S} =< N_v, N_o, \boldsymbol{E}, N_f, D_o >$: number of vehicle, number of order, entropy, number of fleet management vehicle, order distribution

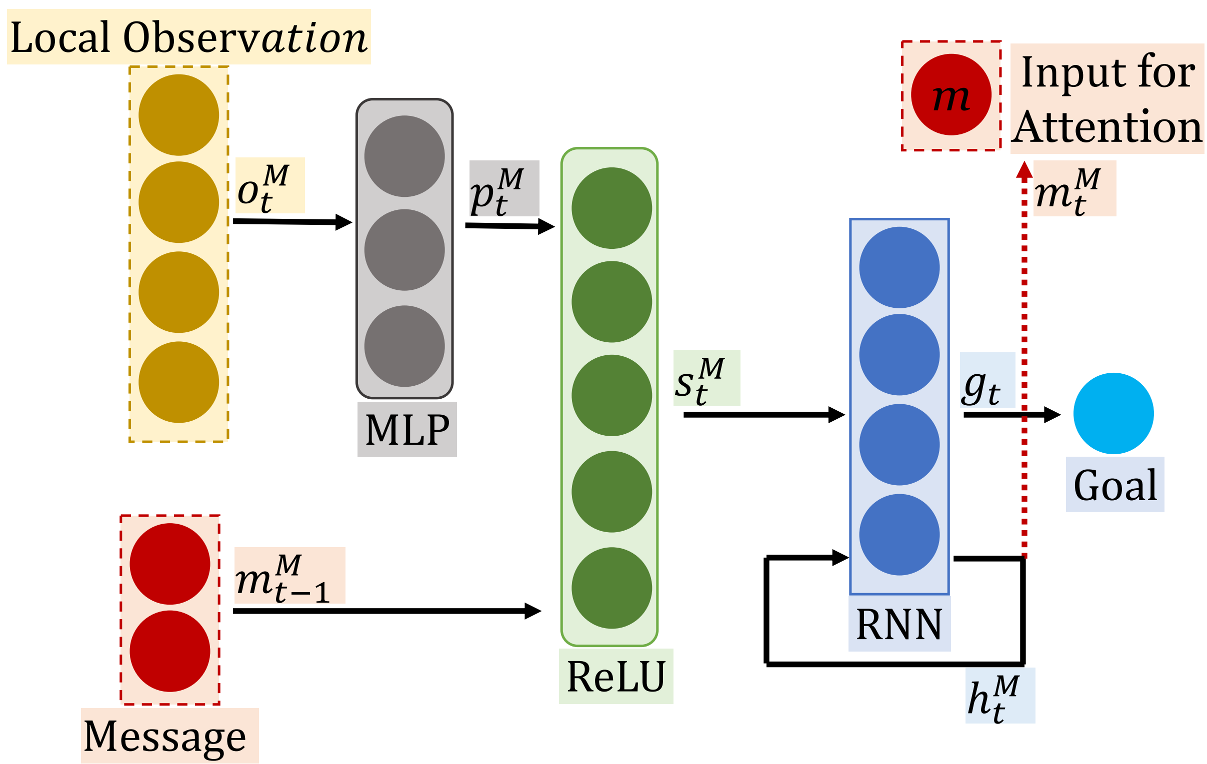
Component Modules

- State $s_t \in \mathcal{S}$, Observation $o_t \in \mathcal{O}$:
  - each agent i draws private observation o_t^i correlated with s_t
  - observation of *manager* is joint observation of its *worker*s
  - $\mathcal{S} =< N_v, N_o, \boldsymbol{E}, N_f, D_o >$: number of vehicle, number of order, entropy, number of fleet management vehicle, order distribution

- Action $a_t \in \mathcal{A}$:
  - *manager*: generate g_t as action, as a direction for *worker*s to follow
  - *worker:* generate ranking weight ω_t as action, as an unified formulation for order dispatching and fleet management

Component Modules

• S



py,

ion

•

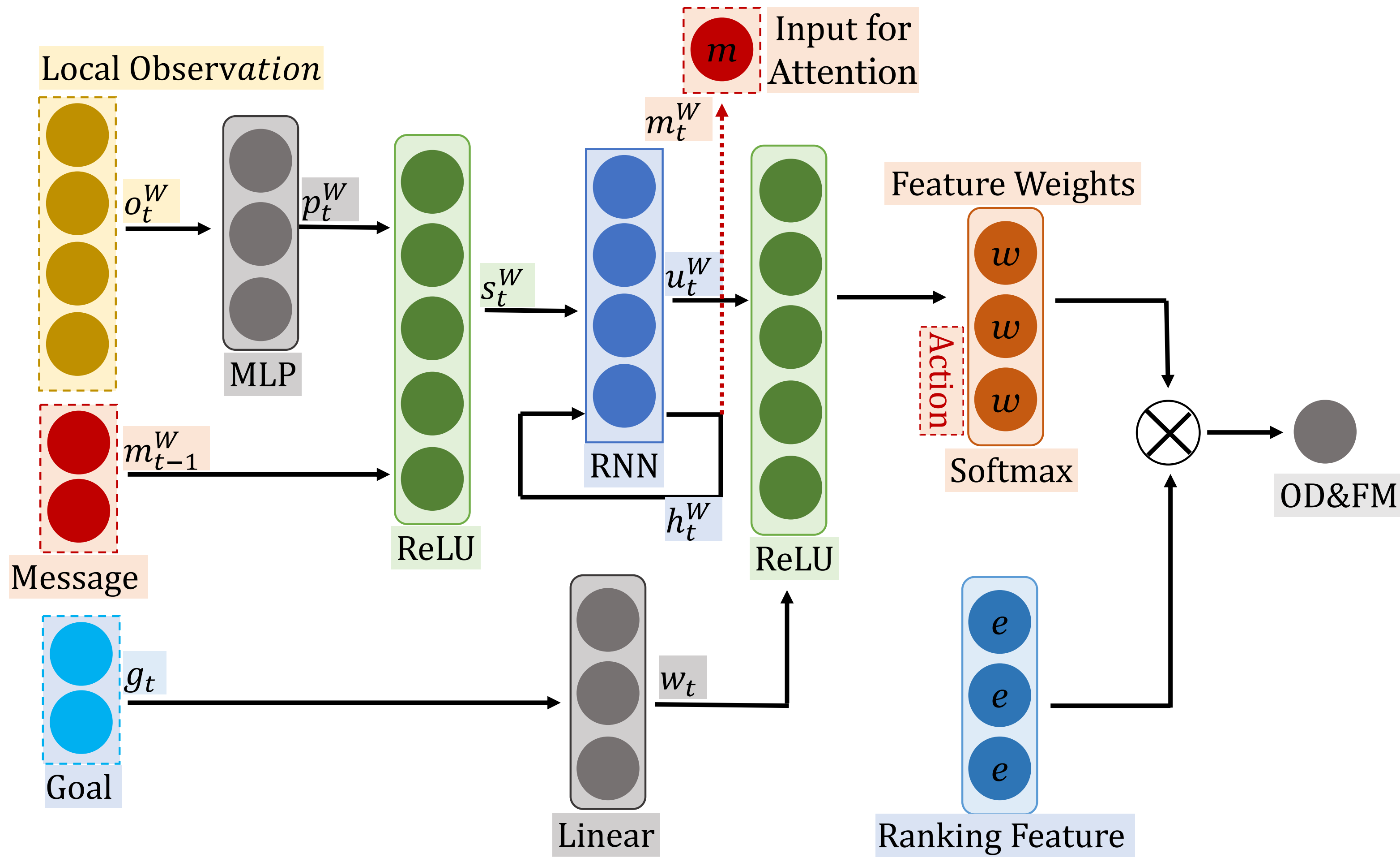
Component Modules

- State $s_t \in \mathcal{S}$, Observation $o_t \in \mathcal{O}$:
  - each agent i draws private observation o_t^i correlated with s_t
  - observation of *manager* is joint observation of its *worker*s
  - $\mathcal{S} =< N_v, N_o, \boldsymbol{E}, N_f, D_o >$: number of vehicle, number of order, entropy, number of fleet management vehicle, order distribution

- Action $a_t \in \mathcal{A}$:
  - *manager*: generate g_t as action, as a direction for *worker*s to follow
  - *worker:* generate ranking weight ω_t as action, as an unified formulation for order dispatching and fleet management

Component Modules

- S



py,

-

ion

Component Modules

- State $s_t \in \mathcal{S}$, Observation $o_t \in \mathcal{O}$:
  - each agent i draws private observation o_t^i correlated with s_t
  - observation of *manager* is joint observation of its *worker*s
  - $\mathcal{S} =< N_v, N_o, \boldsymbol{E}, N_f, D_o >$: number of vehicle, number of order, entropy, number of fleet management vehicle, order distribution

- Action $a_t \in \mathcal{A}$:
  - *manager*: generate g_t as action, as a direction for *worker*s to follow
  - *worker:* generate ranking weight ω_t as action, as an unified formulation for order dispatching and fleet management

Component Modules

- State $s_t \in \mathcal{S}$, Observation $o_t \in \mathcal{O}$:
  - each agent i draws private observation o_t^i correlated with s_t
  - observation of *manager* is joint observation of its *worker*s
  - $\mathcal{S} =< N_v, N_o, \boldsymbol{E}, N_f, D_o >$: number of vehicle, number of order, entropy, number of fleet management vehicle, order distribution
- Action $a_t \in \mathcal{A}$:
  - *manager*: generate g_t as action, as a direction for *worker*s to follow
  - *worker:* generate ranking weight ω_t as action, as an unified formulation for order dispatching and fleet management
- Reward $\mathcal{R}$:
  - *manager*: extrinsic reward from environment
  - *worker*: intrinsic reward from *manager*

• St

ADI: accumulated driver income; **ORR**: order response rate

•

•

•

y,

$$\mathcal{R}_t^W = \frac{1}{c}\sum \underline{d_{cos}}(s_t - s_{t-i}, g_{t-i})$$

$$d_{cos}(\alpha, \beta) = \alpha^T \beta / (|\alpha| \cdot |\beta|)$$

• A

$$\mathcal{R}_t^M = \mathcal{R}_{ADI} + \mathcal{R}_{ORR}$$

n

• R

$$\mathcal{R}_{ORR} = \underline{\sum_{grid} (E - \bar{E})^2 + \sum_{area} D_{KL}(\mathcal{P}_t^o || \mathcal{P}_t^v)}$$

Former: *global* optimization; Later: *local* optimization

Component Modules

- State $s_t \in \mathcal{S}$, Observation $o_t \in \mathcal{O}$:
  - each agent i draws private observation o_t^i correlated with s_t
  - observation of *manager* is joint observation of its *worker*s
  - $\mathcal{S} =< N_v, N_o, \boldsymbol{E}, N_f, D_o >$: number of vehicle, number of order, entropy, number of fleet management vehicle, order distribution

- Action $a_t \in \mathcal{A}$:
  - *manager*: generate g_t as action, as a direction for *worker*s to follow
  - *worker:* generate ranking weight ω_t as action, as an unified formulation for order dispatching and fleet management

- Reward $\mathcal{R}$:
  - *manager*: extrinsic reward from environment
  - *worker*: intrinsic reward from *manager*

Hierarchical Multi-agent Reinforcement Learning

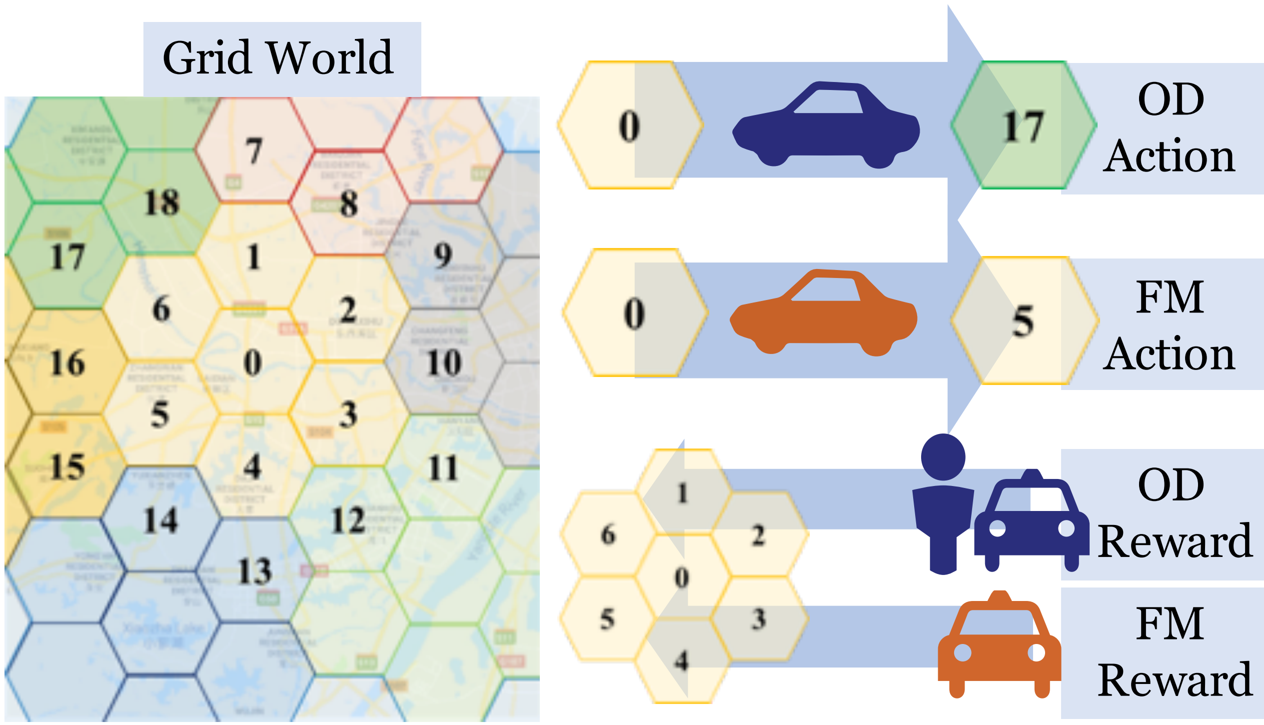


Figure: Hierarchical Multi-agent Reinforcement Learning

Content

- Problem Background
  - Ride-hailing Platform
  - Multi-scale Joint Ride-hailing Platform
- Architecture
  - Hierarchical Multi-agent Reinforcement Learning
- **Experiment**
  - Simulator
  - Real-world Data Experiments
  - Synthetic Data Experiments
- Conclusion

Simulator

- Simulator setting

  grid simulator proposed by [Kaixiang Lin, et al.] evaluated by online A/B test in terms of GMV

- Real-world data

  datasets contain order information (e.g. order price, origin, destination) and trajectories (e.g. positions, status) of vehicles in three large cities (182, 126, 112 grids)

- Synthetic data

  datasets are built via sampling real-world datasets, where uptown and downtown differ through different *sampling rate*

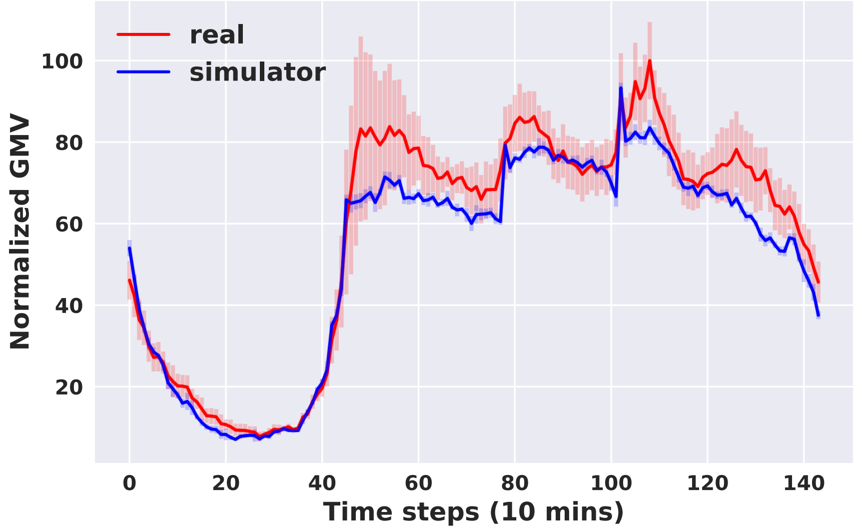


Figure: Simulator

Simulator Link

Data Link

Ref: Kaixiang Lin, et al. Efficient Large-Scale Fleet Management via Multi-Agent Deep Reinforcement Learning. KDD, 2018.

Real-world Data Experiments

ADI: accumulated driver income; **ORR**: order response rate

Table 1: Performance comparison of competing methods in terms of ADI and ORR with respect to the performance of RAN. For a fair comparison, the random seeds that control the dynamics of the environment are set to be the same across all methods.

City	City A		City B		City C	
Metrics	Normalized ADI	Normalized ORR	Normalized ADI	Normalized ORR	Normalized ADI	Normalized ORR
DQN	+5.71% ± 0.02%	+2.67% ± 0.01%	+6.30% ± 0.01%	+3.01% ± 0.02%	+6.11% ± 0.02%	+3.04% ± 0.01%
MDP	+7.11% ± 0.05%	+2.71% ± 0.03%	+7.89% ± 0.05%	+3.13% ± 0.04%	+7.53% ± 0.03%	+3.19% ± 0.03%
DDQN	+6.68% ± 0.04%	+3.19% ± 0.04%	+7.75% ± 0.06%	+4.06% ± 0.05%	+7.62% ± 0.04%	+4.58% ± 0.05%
MFOD	+6.62% ± 0.03%	+3.71% ± 0.02%	+7.91% ± 0.04%	+4.01% ± 0.02%	+7.32% ± 0.02%	+4.60% ± 0.01%
CoRide-	+9.27% ± 0.04%	+4.23% ± 0.03%	+8.73% ± 0.03%	+4.35% ± 0.02%	+9.06% ± 0.03%	+4.23% ± 0.04%
CoRide	+**9.80**% ± 0.04%	+4.81% ± 0.05%	+**8.94**% ± 0.06%	+4.89% ± 0.04%	+**9.23**% ± 0.05%	+5.19% ± 0.04%

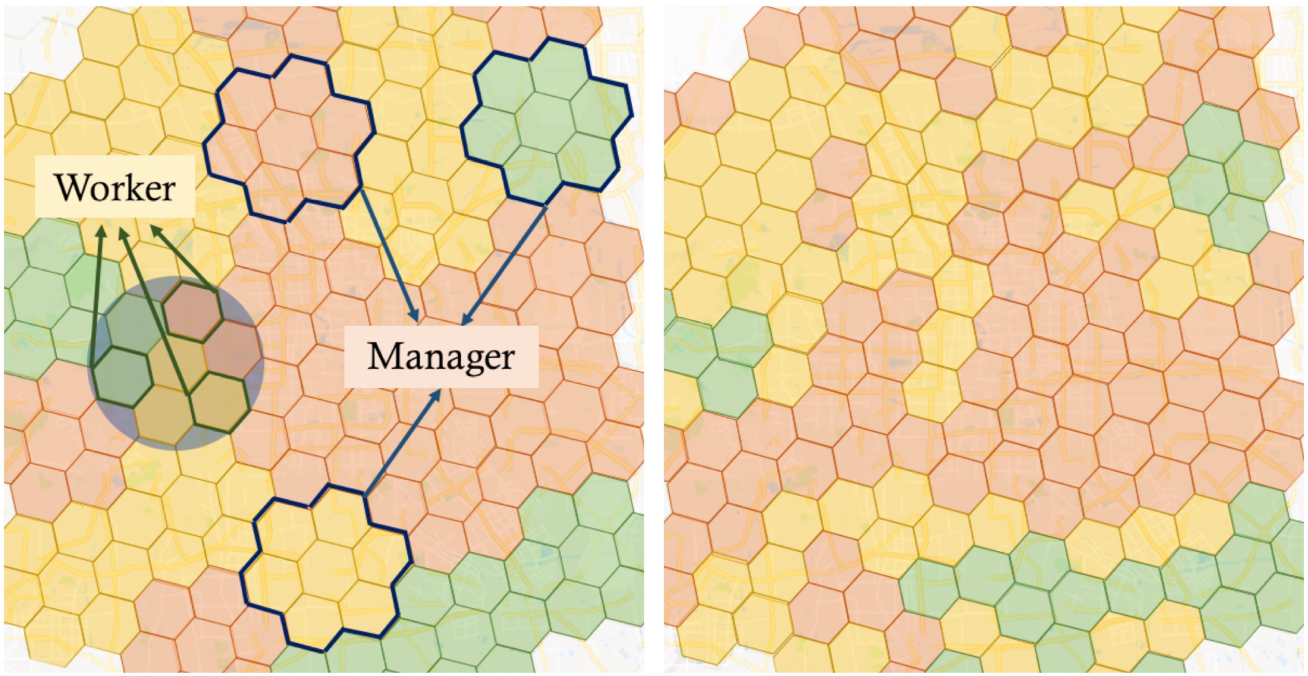
Baselines

[RAN] A random dispatching algorithm considering no additional information.
[DQN] [Minne Li, et al.] conducted action-value function approximation based on Q-network.
[MDP] [Zhe Xu, et al.] implemented dispatching through a learning and planning approach.
[DDQN] [Zhaodong Wang, et al.] introduced a double-DQN with spatial-temporal action search.
[MFOD] [Minne Li, et al.] modeled order dispatching with mean field reinforcement learning.
[CoRide-] CoRide without multi-head attention mechanism.

Ref: Zhaodong Wang, et al. Deep Reinforcement Learning with Knowledge Transfer for Online Rides Order Dispatching. ICDM, 2018.
Ref: Minne Li, et al. Efficient Ride-sharing Order Dispatching with Mean Field Multi-agent Reinforcement Learning. WWW, 2019.
Ref: Zhe Xu, et al. Large Scale Order Dispatch in On Demand Ride-Hailing Platforms: A Learning and Planning Approach. KDD, 2018.

Real-world Data Experiments

grids with more orders or higher attention value are shown in red (green if opposite) and the gap is proportional to the shade of colors



(a) Sampled Attention Value (b) Demand-supply Gap

Figure: Sampling attention value and demand-supply gap

Synthetic Data Experiments

AST: accumulated on-service time; **TNF**: total number of finished order

Table 2: Performance comparison of competing methods in terms of AST and TNF with three different *discounted rates* (DR). The numbers in Trajectory denote gridID in Figure 7 and its color denotes the district it located in. O and W mean the vehicle is On-service and Waiting at the current grid. Also, we use underlined number to present fleet management.

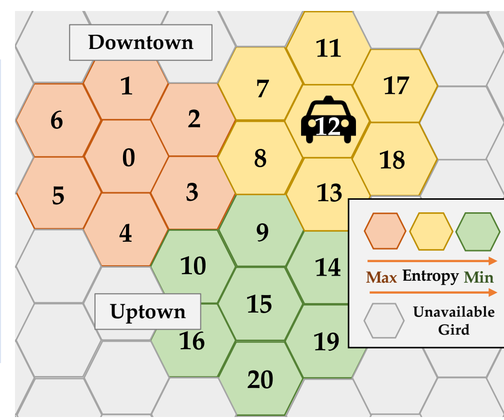
DR	20%			30%			40%		
Metrics	Trajectory	AST	TNF	Trajectory	AST	TNF	Trajectory	AST	TNF
RES	13,9,W,14,W,13,8,2,7,11	8	8	13,W,14,W,W,13,8,W,7,11	6	6	13,W,14,W,W,W,W,19,O,9	5	4
REV	O,O,15,W,20,O,O,O,O,11	9	3	O,O,15,W,W,W,O,O,O,11	7	2	O,O,15,W,W,W,20,W,O,14	6	3
CoRide	13,9,W,O,0,4,O,2,O,5	**9**	**6**	13,W,O,11,W,W,O,O,0,5	7	4	13,W,W,O,17,W,W,O,0,3	6	3
CoRide+	13,9,W,O,0,4,O,2,O,5	**9**	**6**	8,3,0,2,O,4,O,2,O,5	**8**	**5**	8,3,0,2,O,4,O,2,O,5	**8**	**5**

Baselines

[RES] This response-based method aims to achieve higher TNF (ORR). Order with short duration will gain high priority.

[REV] The revenue-based algorithm focus on higher AST (ADI) via assigning vehicles to high price orders.

[CoRide+] CoRide in joint order dispatching and fleet management.



Content

- Problem Background
  - Ride-hailing Platform
  - Multi-scale Joint Ride-hailing Platform
- Architecture
  - Hierarchical Multi-agent Reinforcement Learning
- Experiment
  - Simulator
  - Real-world Data Experiments
  - Synthetic Data Experiments
- **Conclusion**

Conclusion

- Hierarchical setting
  adapt to **dynamic** environment by *manager* setting abstract directions to *workers*
- Joint setting
  fully exploit precise driver source
- Multi-agent setting
  parallel implementation & collaboration in both *manager* and *worker* layers
- Multi-scale setting
  capture demand-supply dynamics from both district and grid **scale**

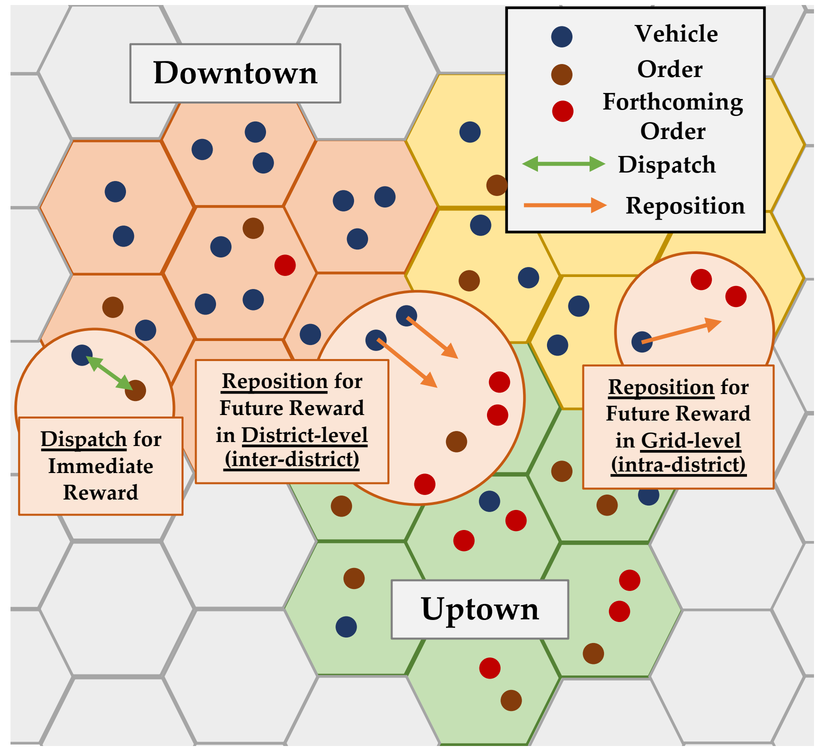


Figure: CoRide

Poster Session II: 6:30pm-9:00pm, Nov 6

Workshop on Artificial Intelligence in Transportation: 2:00pm-5:00pm, Nov 7

Thanks for Your Listening

Slides Link

Jerry's Wechat

Full Paper Link

Email: jinjiarui97@gmail.com

Website: https://jinjiarui.github.io/